

# GRANT NEILSON

Bioinformatician with six years of industry experience analyzing multiple NGS data types, with a specialism in whole genome sequencing (WGS). Highly proficient in R and Python, with a strong track record of developing analytical tools and packages for high-dimensional omics data. Passionate about building robust analytical workflows and contributing to open-source repositories to drive forward scientific research. An articulate and reliable individual with excellent communication skills and a strong track record of effective collaboration across multidisciplinary teams.

View this CV online with links at <https://grantn5.github.io/cv/>

## CONTACT

✉ [grant.neilson5@gmail.com](mailto:grant.neilson5@gmail.com)

🐙 [GitHub](#)

in [LinkedIn](#)

## INDUSTRY EXPERIENCE

2026  
|  
2023

### Senior Computational Biologist

[CoSyne Therapeutics](#)

📍 London, UK

- Drove biomarker discovery in glioblastoma by conducting WGS analysis on 80 primary tumour-derived cell lines, delivering comprehensive genetic characterisation (SNVs, SVs, CNVs). Integrated genomic profiles with CRISPRi screen data using advanced statistical models to uncover novel genetic dependencies and biomarker-target pairs.
- Designed and implemented the end-to-end bioinformatic analysis for high-throughput single-cell Mosaic drug screening, deploying hyperscale cloud infrastructure to run alignment and quality-control pipelines for ~3 million cells. Designed and implemented scalable pipelines for cell demultiplexing, sensitivity profiling of cell lines, and multi-omic biomarker identification.
- Executed single-cell CNV (sc-CNV) calling on proprietary perturb scRNA-seq data, identifying sub-clonal populations and contributing to the target identification platform.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.
- Internal R&D: Responsible for developing an internal container system using Docker and ECR (later handed over to engineering). Author and Lead developer on several internal R and Python packages with responsibilities for maintenance. Established best practices for code development and review. Aided in the development of a project templating system using `cruff`.

2023  
|  
2020

### Computational Biologist

[Fios Genomics](#)

📍 Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Conducted complex regression model analyses for biomarker discovery in bladder and lung cancer.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

## LANGUAGES

📊 R  
🐍 Python  
🔗 Bash  
🔗 Nextflow

## TECHNOLOGIES

🐙 GitLab/GitHub  
📦 Docker  
aws aws  
🖥️ Azure

## OPEN SOURCE & PUBLIC PROJECTS

🐙 [Sarek](#)  
🐙 [bixverse](#)  
🐙 [mainfoldsR](#)

Made with the R package [pagedown](#).

The source code is available on [GitHub](#).

Last updated on 2026-02-12.

2020  
|  
2019

## Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School

📍 Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician. This was unfortunately cancelled due to COVID-19.



## EDUCATION

2019  
|  
2015

### MSCi Biology

University of Southampton

📍 Southampton, UK

- Grade : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015  
|  
2013

### School

Merchiston Castle School

📍 Edinburgh, UK

- A Levels: Biology, Chemistry, Maths



## SELECTED PUBLICATIONS

2025

### Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, Grant Neilson, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preprocessing
- doi: <https://doi.org/10.1101/2025.04.17.649387>

2025

### Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, Grant Neilson, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preprocessing
- doi: <https://doi.org/10.1101/2025.04.17.649395>

2021

• **Novel epigenetic clock for fetal brain development predicts prenatal age for cellular stem cell models and derived neurons**

Molecular Brain

- Leonard C. Steg, Gemma L. Shireby, Jennifer Imm, Jonathan P. Davies, Alice Franklin, Robert Flynn, Seema C. Namboori, Akshay Bhinge, Aaron R. Jeffries, Joe Burrage, Grant W. A. Neilson, Emma M. Walker, Leo W. Perfect, Jack Price, Grainne McAlonan, Deepak P. Srivastava, Nicholas J. Bray, Emma L. Cope, Kimberley M. Jones, Nicholas D. Allen, Ehsan Pishva, Emma L. Dempster, Katie Lunnon, Jonathan Mill & Eilis Hannon

2020

• **Recalibrating the epigenetic clock: implications for assessing biological age in the human cortex**

Brain

- Gemma L Shireby, Jonathan P Davies, Paul T Francis, Joe Burrage, Emma M Walker, Grant W A Neilson, Aisha Dahir, Alan J Thomas, Seth Love, Rebecca G Smith, Katie Lunnon, Meena Kumari, Leonard C Schalkwyk, Kevin Morgan, Keeley Brookes, Eilis Hannon, Jonathan Mill
- Role: Data Analysis and Alignment